# DATA100 Midterm 1

Tuesday, October 6, 2026

**Instructions**: Put your name, WLU id, **and section** on the answer sheet (which is separate).

**Answer all questions on the answer sheet.**

Only the **answer sheet** will be graded. No answers on this booklet will be considered. You may use this booklet for scrap paper.

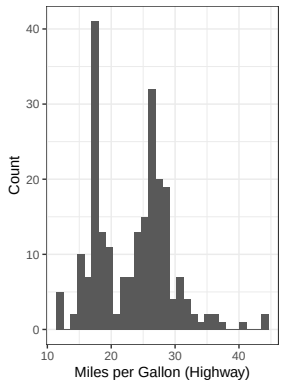
There are 8 pages, with questions printed on the front and back of each sheet.

## Multiple Choice Questions

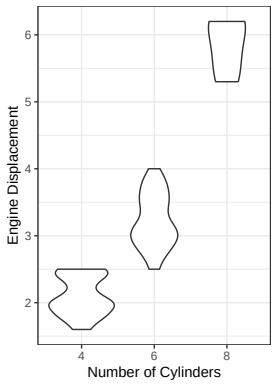
Each question below is worth one mark. Answer on the **answer sheet** provided.

1. Which of the following plots provides information about the kinds of cars that have the highest miles per gallon (highway)?

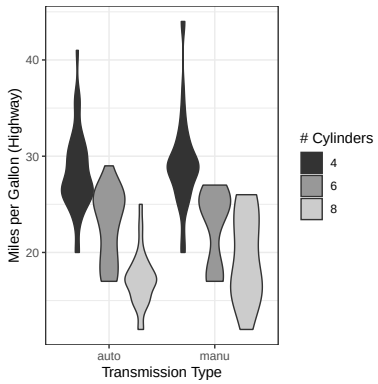
1.a



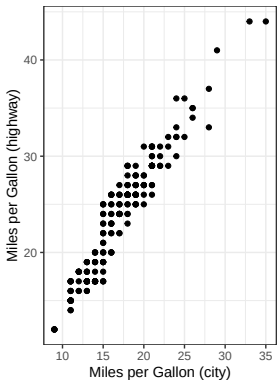

1.b

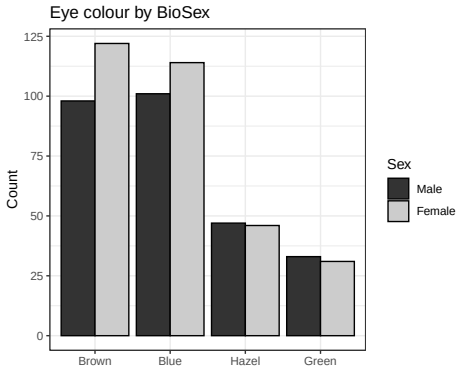



1.c




1.d




2. Which of the following is **NOT** a valid interpretation of the plot on the right?

   a. There is an overall decreasing trend when we move from brown to green eyes.
   b. There were more female subjects than male subjects
   c. There were more female subjects with blue and brown eyes than male subjects
   d. Brown and blue eyes appear to be overall more common than hazel and green

3. Consider subset of data on the right (showing 7 out of 67 rows), collected on people aged 19 and 20 and given one of three drugs to cure their disease. Which of the following plots would provide meaningful information?

   a. A bar plot of `id` versus `name`
   b. A scatterplot of `age` versus `id`
   c. A violin plot of `age`, coloured by `cured`
   d. A bar plot of `cured`, coloured by `medicine`

| id | name | medicine | age | cured |
|---|---|---|---|---|
| 1 | Bonk | A | 19 | yes |
| 2 | Lilah | B | NA | NA |
| 3 | June | C | 19 | no |
| 4 | Judy | B | 19 | yes |
| 5 | Julie | C | 19 | no |
| 6 | Jodie | A | 20 | yes |
| 7 | Jill | A | 20 | no |

4. In `ggplot2`, what is the best description of a `geom`?

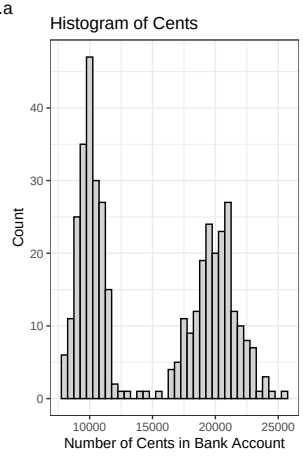
    a. Declaring the way a data set will be used, e.g. by assigning columns to a dataframe.
    b. The way data are assigned to elements of a plot, e.g. assigning `mpg` to the y axis or assigning `cyl` to the colour scheme.
    c. The way elements of a plot are actually drawn, e.g. by declaring that a violin plot will be used.
    d. The way additional plot elements, such as arrows or text, are added tp the plot.

5. Consider a data set which records the number of cents in each person's bank account, where each bank account can only have a whole number of cents (no decimals). What type of variable is this?

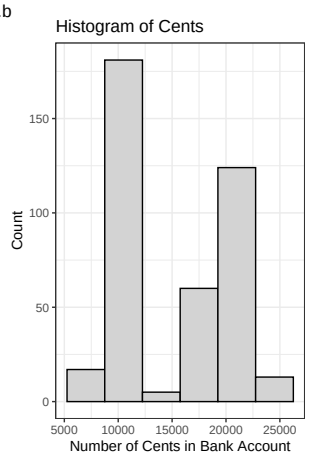
    a. Qualitative - Categorical
    b. Qualitative - Ordinal
    c. Quantitative - Continuous
    d. Quantitative - Discrete

6. Consider the data from Question 5; the number of cents in a person's bank account. Which of the following visualization provides the most useful information?

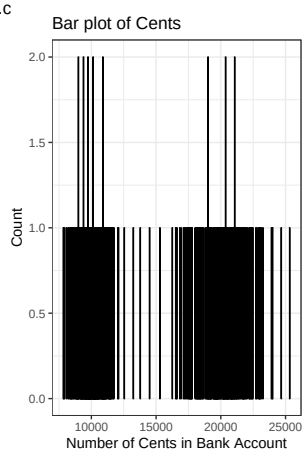
6.a



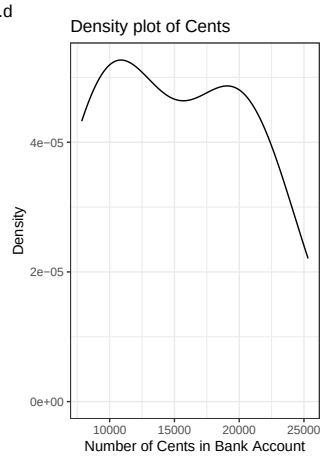

6.b

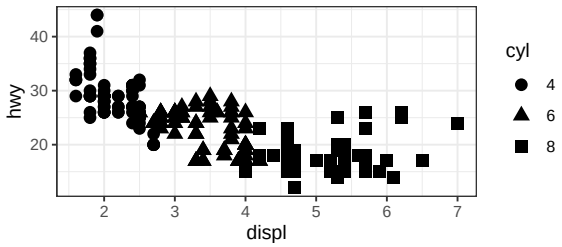



6.c




6.d




Consider the plot on the right for the next question. Note that the name of the data set is `mpg`, with columns `class` for the type of car (sedan, SUV, truck, etc.), `displ` for the size of the engine (displacement) in cubic feet, `hwy` for highway fuel efficiency in miles per gallon, and `cyl` for number of cylinders in the engine.

7. Which code produced the figure above?

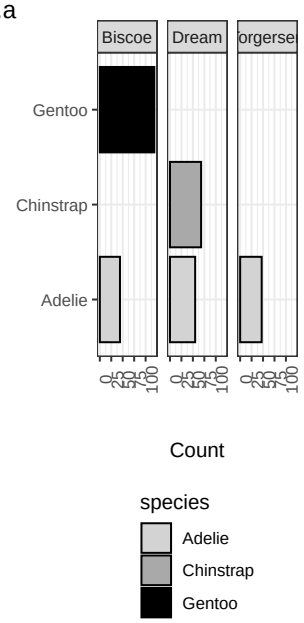
    a. `ggplot(mpg) + aes(x = displ, y = hwy, shape = cyl) + geom_point()`
    b. `ggplot(mpg) + aes(x = hwy, y = displ, colour = cyl) + geom_point()`

c. `ggplot(mpg) + aes(x = displ, y = hwy, colour = cyl) + geom_scatterplot()`
d. `ggplot(mpg) + geom_scatterplot(x = hwy, y = displ, shape = cyl)`

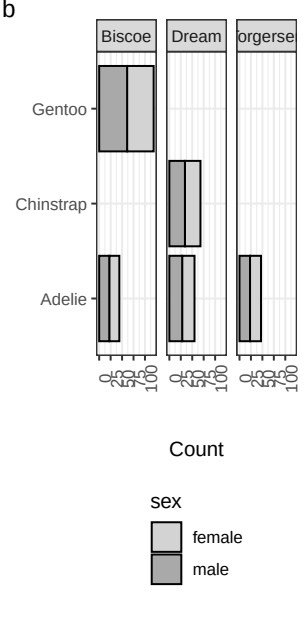
8. Consider the following code. Which figure would it produce?

```
ggplot(penguins) +
    aes(x = species, fill = species) +
    geom_bar() +
    facet_wrap(~ island) +
    coord_flip() +
    labs(
        x = NULL, y = "Count"
    )
```

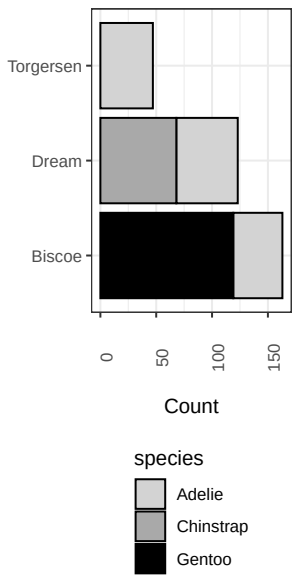
8.a



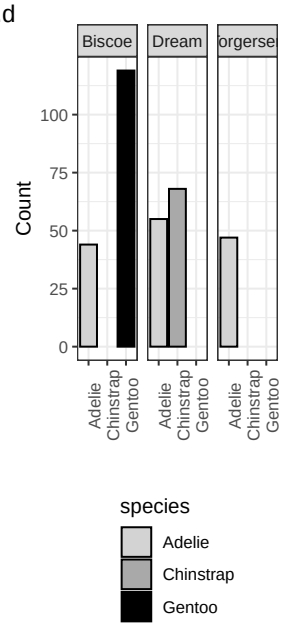

8.b




8.c




8.d




9. Consider the snippet of data on the right (8 out of 200 rows). Which of the following plots would best answer the question "Do students in different programs tend to score differently?"

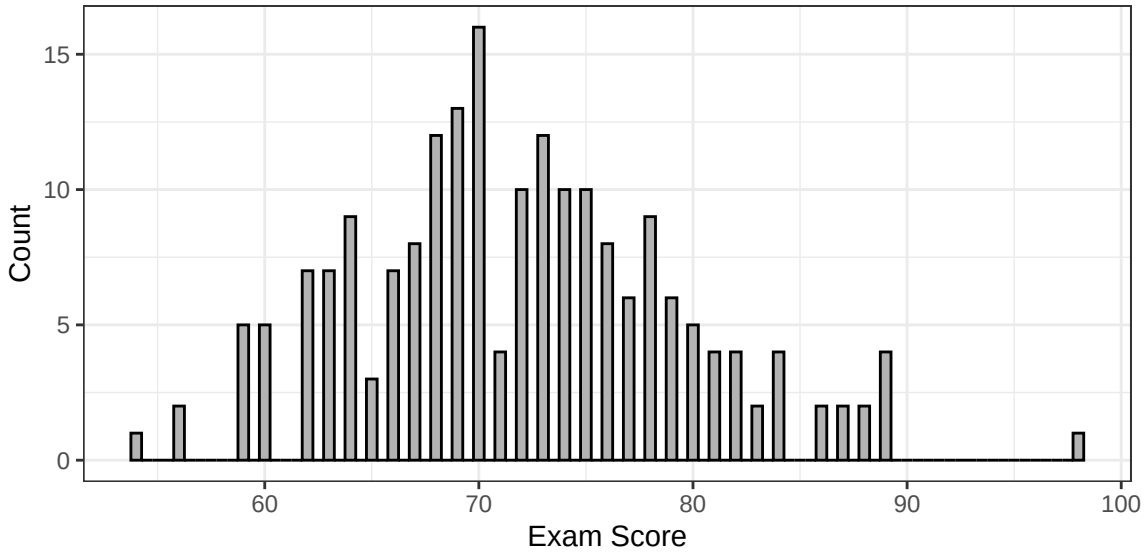
    a. A scatterplot of `hours_studied` versus `exam_score`
    b. A boxplot of `exam_score` for each `program`
    c. A histogram of `program` and `exam_score`
    d. A bar plot of `program` versus `hours_studied`

| id | program | hours_studied | exam_score |
|---|---|---|---|
| 101 | Biology | 6 | 78 |
| 102 | Music | 3 | 65 |
| 103 | Economics | 8 | 88 |
| 104 | Biology | 4 | 70 |
| 105 | Economics | 5 | 74 |
| 106 | Music | 7 | 82 |
| 107 | Biology | 9 | 91 |
| 108 | Economics | 2 | 60 |

10. The `penguins` data set has columns `species` (three species), `island` (three islands), `sex`, and `body_mass_g` (body mass in grams, a continuous variable). Which of the following mappings is appropriate?

    a. `shape` mapped to `island`
    b. `size` mapped to `species`
    c. `shape` mapped to `body_mass_g`
    d. `alpha` mapped to `island`

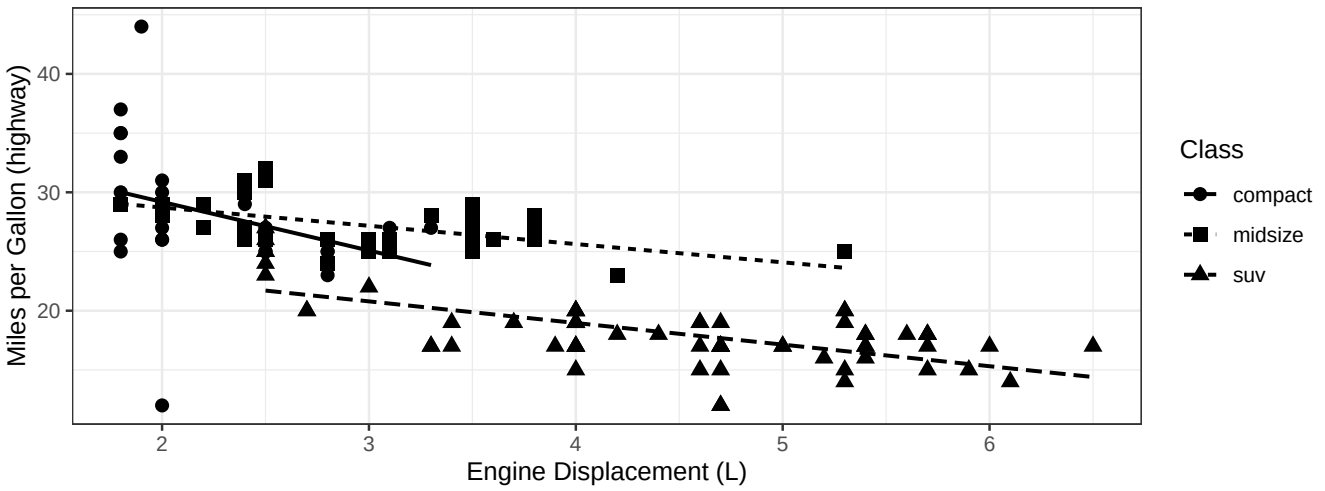
Consider the histogram below for the next question. It shows exam scores for 200 students.




11. Every second bar in the histogram above is empty. What is the most likely explanation?

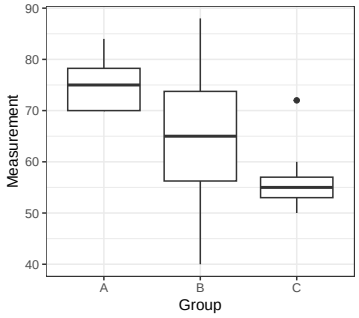
    a. The scores were recorded as whole numbers, and the binwidth is smaller than that
    b. The data have two groups of students with alternating score patterns
    c. The binwidth is too large to show any detail
    d. Half of the scores are missing from the data set

Consider the plot below for the next question. It shows highway fuel efficiency against engine displacement for three classes of vehicle, with a straight line fit to each class. Note that the name of the data set is `mpg`, with columns `class` for the type of car, `displ` for the size of the engine (displacement) in litres, and `hwy` for highway fuel efficiency in miles per gallon.




12. Which of the following conclusions is supported by the plot above?

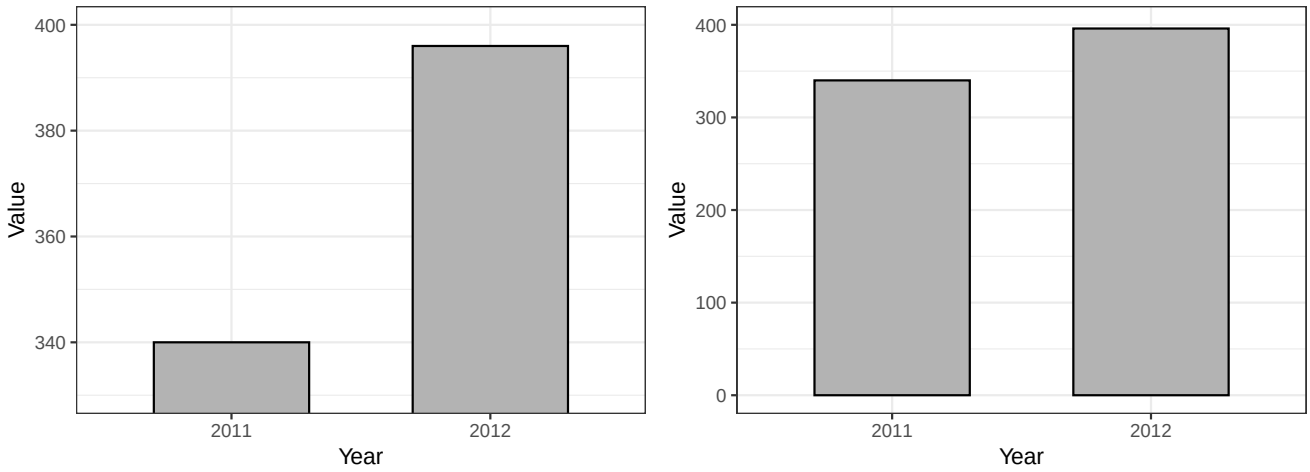
    a. Larger engines directly cause lower fuel efficiency.
    b. Within each class, highway mpg tends to fall as displacement rises.
    c. Vehicle class has no relationship with highway mpg.
    d. The point far below the compact line must be a data entry error.




13. Consider the plot on the right, showing the boxplots of a measurement for three groups. Which of the following is **NOT** a valid interpretation?

    a. Group C has the smallest range of the three groups.
    b. The observations in Group B are spread roughly symmetrically around its median.
    c. About three quarters of the observations in Group B are below the median of Group A.
    d. Group C has fewer observations than Group A.

Consider the two bar charts below for the next question. Both show the same two values, 340 in 2011 and 396 in 2012.




14. Which statement about the two charts is correct?

    a. The left chart is better because the increase is easier to see.
    b. The left chart exaggerates the increase because the bar lengths are no longer proportional to the values.
    c. Both charts are equally good because they show the same data.
    d. The right chart hides the increase and should not be used.

15. Which visual cue do people judge most accurately?

    a. The angle of a pie slice
    b. The area of a circle
    c. The shade of grey

d. Position along a common scale

## Short Answer Questions

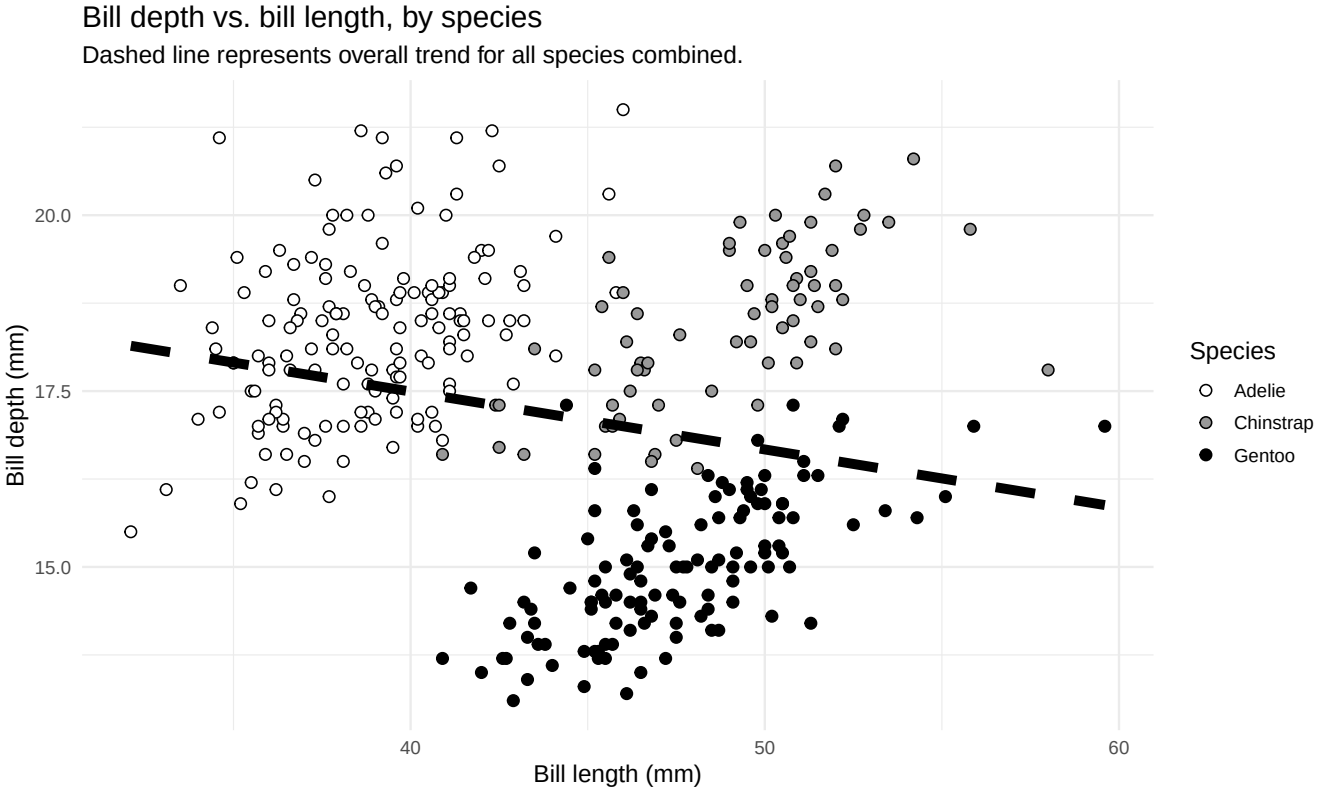
16. Plot the following data on the axes provided in the answer sheet. Choose whichever plot you believe is most appropriate.

    - [2 marks] The plot type is appropriate.
    - [1 mark] The data are plotted reasonably accurately.
    - [1 mark] Appropriate labels are used.
    - [1 mark] Colours / shapes / fills are used appropriately (for colour, use different levels of shading; the scans will be in black and white so they won't pick up different colours).

| student_number | assessment | grade |
|---|---|---|
| 123456789 | A1 | 90 |
| 123456789 | A2 | 83 |
| 123456789 | A3 | 74 |
| 123456789 | MT | 89 |
| 124624353 | A1 | 60 |
| 124624353 | A2 | 65 |
| 124624353 | MT | 76 |
| 124624353 | A3 | 70 |
| 984367234 | A1 | 73 |
| 984367234 | MT | 71 |
| 984367234 | A2 | 74 |
| 984367234 | A3 | 76 |

17. The plot below shows bill depth against bill length for the penguins data set, with a different shade of grey for each of the three species of penguin: `Adelie`, `Chinstrap`, and `Gentoo`.

    a. [1 mark] **On the answer sheet**, sketch a trend line for **each** of the three species directly on the plot.
    b. [3 marks] Which species's trend line does the best job summarizing its data? Which species's trend line does the worst job? Explain your reasoning using what you see in the plot.
    c. [3 marks] Suppose you are only shown the dashed overall trend line, and it has a negative slope. Could you conclude that bill depth decreases with bill length within each species? Why or why not?



A library keeps a checkouts data set: one row per checkout, with columns `genre` ("fiction", "nonfiction", "children"), `format` ("print" or "ebook"), `day_of_week`, and `checkout_count` (number of checkouts). You are given the following snippet of code:

```
checkouts |>
    filter(genre != "children") |>
    ggplot(aes(x = day_of_week, y = checkout_count, fill = format)) +
        geom_col(position = "fill") +
        facet_wrap(~genre) +
        labs(
            title = "???",
            x = "Day of the week",
            y = "Proportion of checkouts",
            fill = "Format"
        )
```

18. Use the code above to answer the questions.

    a. [2 marks] What does position = "fill" do to the bars, and what kind of comparison does it let you make that position = "stack" would not?
    b. [2 marks] What relationship is this plot designed to show?
    c. [1 mark] Suppose every bar in the *nonfiction* panel looks close to 50/50 between *print* and *ebook*, on every day of the week. What is the covariation between `day_of_week` and `format` in this case?
    d. [2 marks] If the filter() line were removed, how would the plot change?